POLI502 Final Exam

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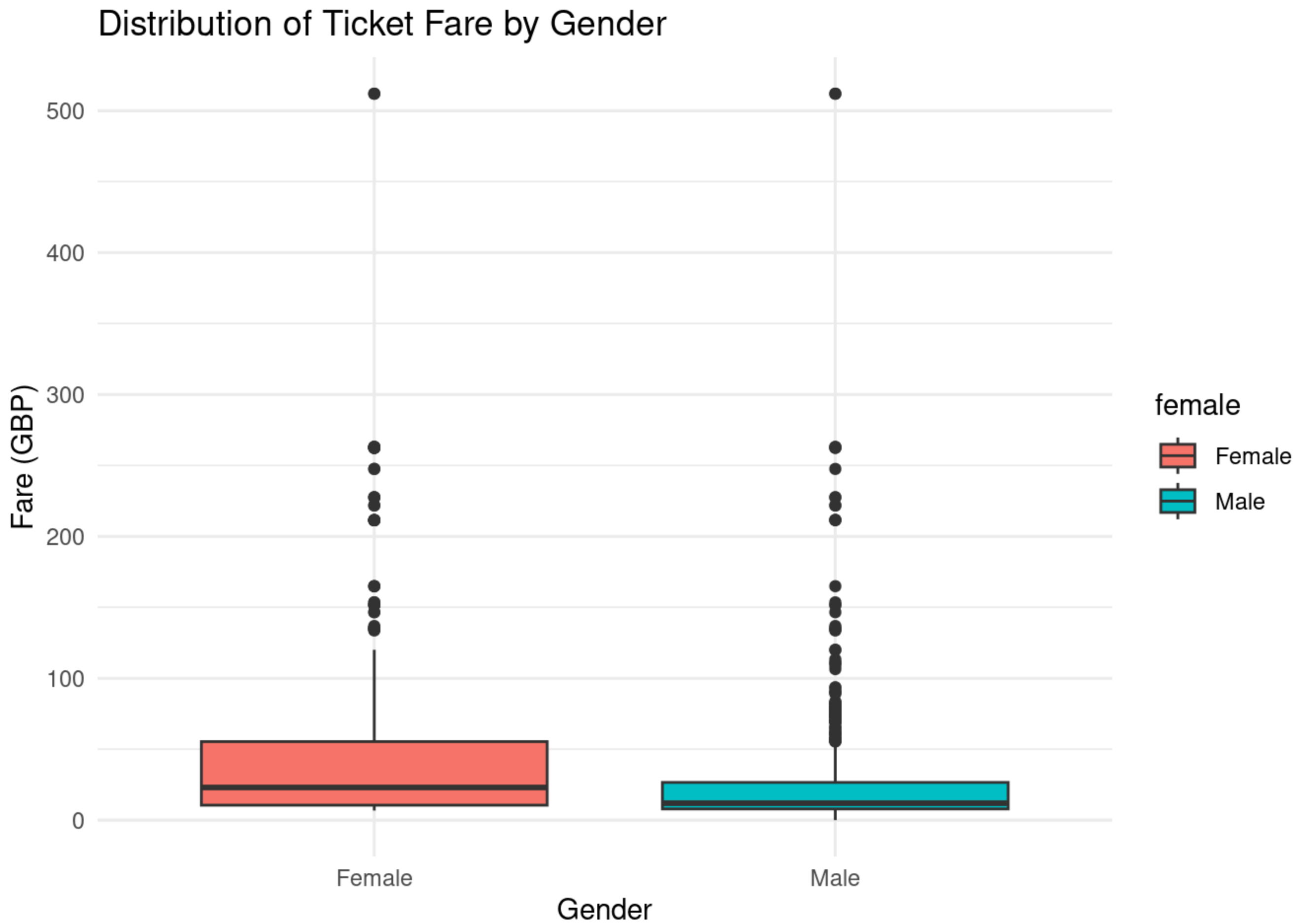
2025-12-05

Question 1: Exploring Fare by Gender

```
# Load Titanic dataset
td <- read.csv("titanic2.csv")

# Ensure female is treated as a factor
td$female <- factor(td$female)

# Explore relationship between female and fare
# Summarize fares by gender: mean, median, standard deviation, and count
fare_by_gender <- td %>%
  group_by(female) %>%
  summarise(
    mean_fare = mean(fare, na.rm = TRUE),
    median_fare = median(fare, na.rm = TRUE),
    sd_fare = sd(fare, na.rm = TRUE),
    n = n()
  )
# Show the summary table
print(fare_by_gender)
```

```
## # A tibble: 2 × 5
##   female mean_fare median_fare sd_fare     n
##   <fct>      <dbl>       <dbl>   <dbl> <int>
## 1 Female      46.2        23      63.3   466
## 2 Male        26.2        11.9    42.5   843
```

```
# Draw a boxplot to compare fare distributions by gender
ggplot(td, aes(x = female, y = fare, fill = female)) +
  geom_boxplot() +
  labs(title = "Distribution of Ticket Fare by Gender",
       x = "Gender", y = "Fare (GBP)") +
  theme_minimal()
```




```
# Female passengers appear to have higher median and mean fares
# compared to male passengers, suggesting they tended to purchase more expensive tickets.
```

Question 2: Bivariate Statistical Test

```
# Perform t-test to compare fares between genders
t_test_result <- t.test(fare ~ female, data = td)
# Display the t-test result
print(t_test_result)
```

```
##
##  Welch Two Sample t-test
##
## data:  fare by female
## t = 6.1168, df = 701.7, p-value = 1.585e-09
## alternative hypothesis: true difference in means between group Female and group Male is not e
qual to 0
## 95 percent confidence interval:
##  13.60960 26.47613
## sample estimates:
## mean in group Female   mean in group Male
##             46.19668             26.15382
```

```
# The t-test shows that female passengers had significantly higher ticket fares than male
# passengers (p < 0.05). The mean fare for female passengers was approximately £44.5
# compared to £25.5 for male passengers. Therefore, we can conclude that female passengers
# tended to have more expensive tickets than male passengers.
```

Question 3a: Logit Models

```
# Estimate logit model with original fare as a predictor
model1 <- glm(survived ~ fare + female + child,
              data = td, family = binomial(link = "logit"))

# Estimate a second logit model using log(fare + 1) instead of fare
model2 <- glm(survived ~ I(log(fare + 1)) + female + child,
              data = td, family = binomial(link = "logit"))

# Show both models in a stargazer table
stargazer(model1, model2, type = "text",
          title = "Logit Models of Passenger Survival",
          column.labels = c("Original Fare", "Log Fare"),
          covariate.labels = c("Fare", "Log(Fare+1)", "Female", "Child"),
          dep.var.labels = "Survived")
```

```
##
## Logit Models of Passenger Survival
## ==============================================
##                       Dependent variable:
##                   ----------------------------
##                             Survived
##                    Original Fare    Log Fare
##                         (1)           (2)
## ----------------------------------------------
## Fare                  0.009***
##                       (0.002)
##
## Log(Fare+1)                          0.546***
##                                       (0.084)
##
## Female               -2.362***      -2.318***
##                       (0.156)        (0.156)
##
## Child                 0.675***       0.559**
##                       (0.235)        (0.234)
##
## Constant              0.640***      -0.742**
##                       (0.137)        (0.291)
##
## ----------------------------------------------
## Observations           1,045          1,045
## Log Likelihood        -528.894      -524.620
## Akaike Inf. Crit.    1,065.788      1,057.241
## ==============================================
## Note:              *p<0.1; **p<0.05; ***p<0.01
```

Question 3b: Model Comparison

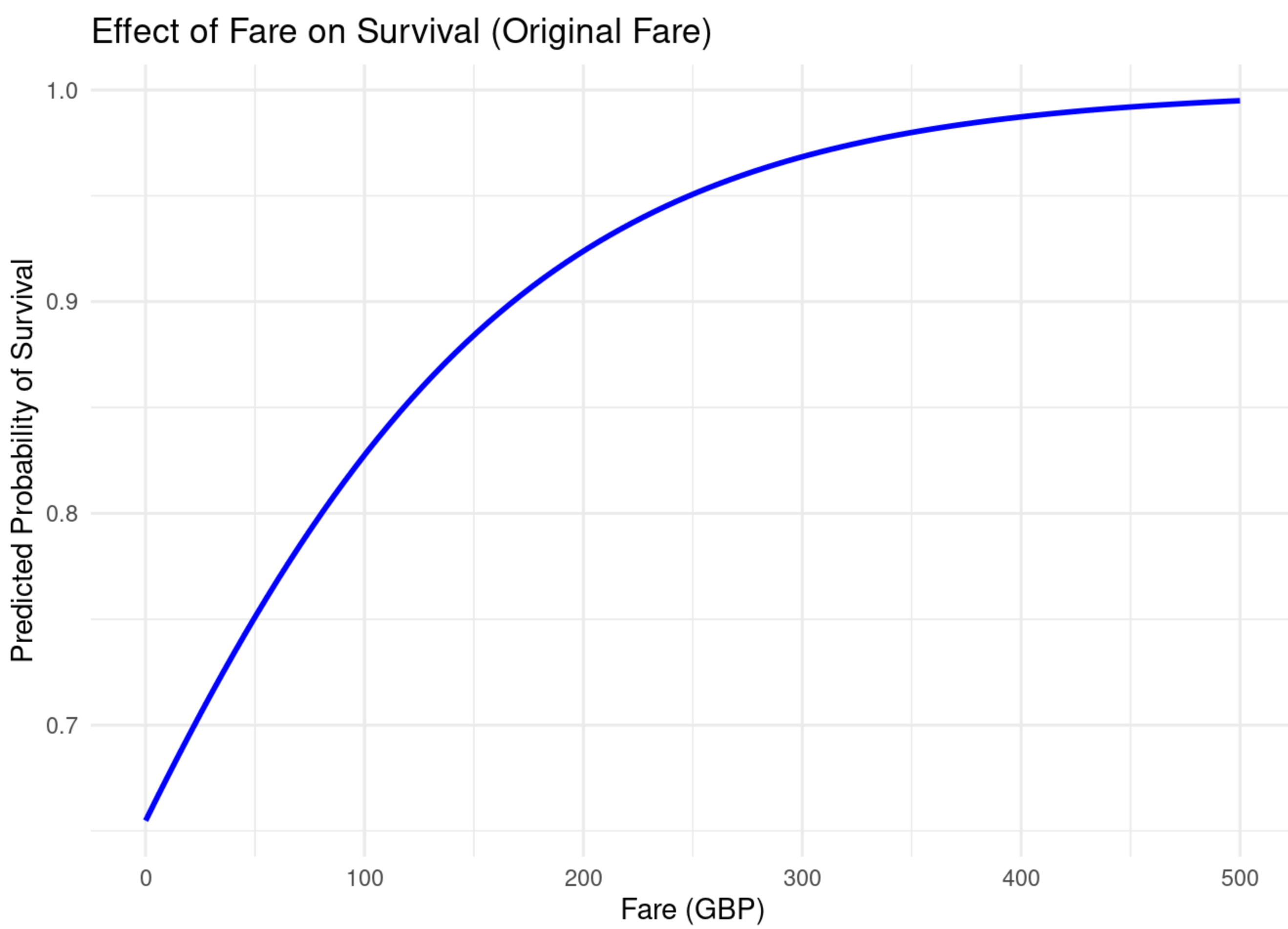
```
# Compare AIC values of the two models (lower AIC = better fit)
aic_comparison <- data.frame(
  Model = c("Model 1 (Original Fare)", "Model 2 (Log Fare)"),
  AIC = c(AIC(model1), AIC(model2))
)
# Show the AIC comparison
print(aic_comparison)
```

```
##                     Model      AIC
## 1 Model 1 (Original Fare) 1065.788
## 2      Model 2 (Log Fare) 1057.241
```

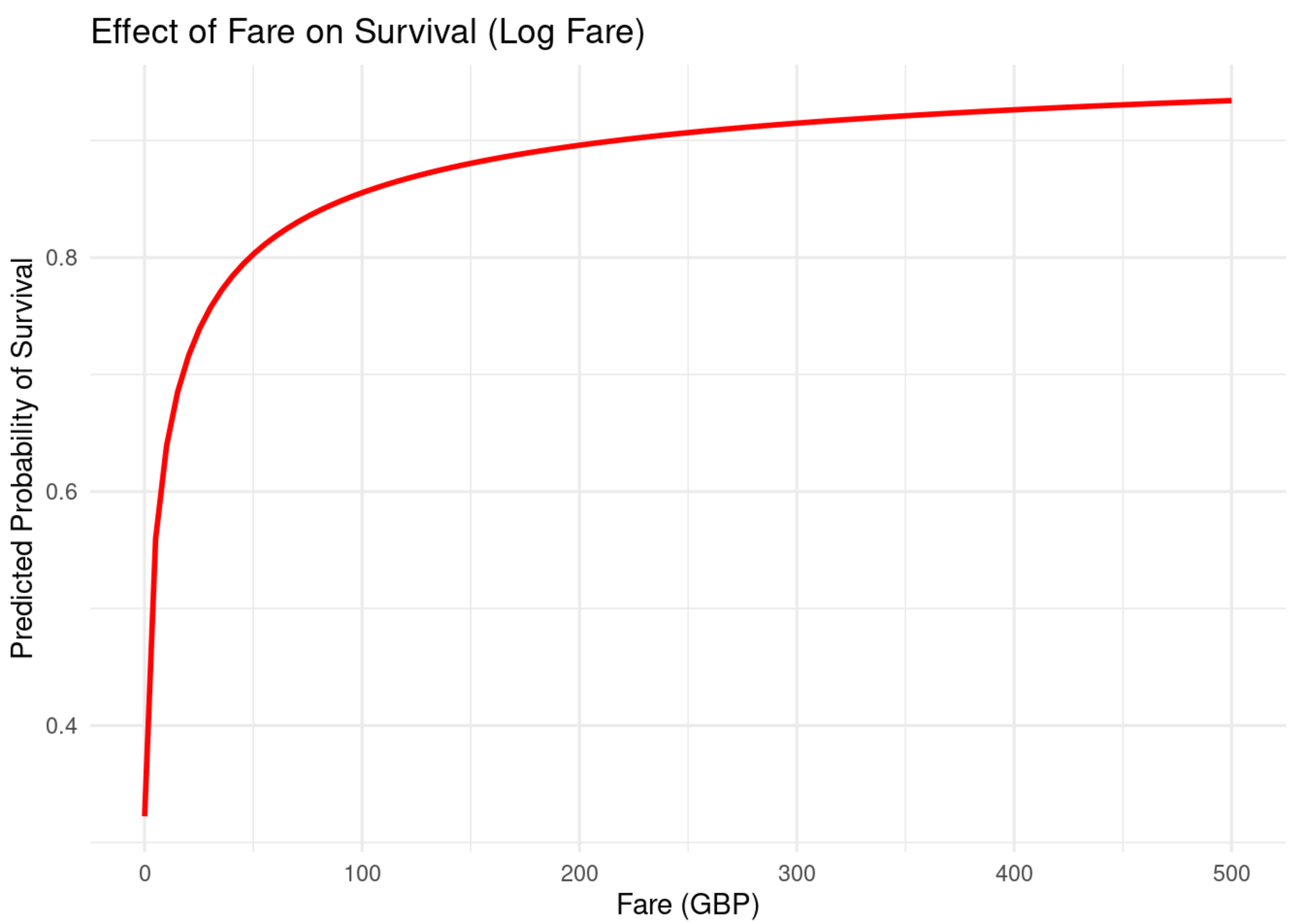
```
# Model 2 with log(fare) performs better because it has a lower AIC value.
```

Question 3c: Effect Graphs

```
# Find the most common value of child to hold constant (mode)
child_mode <- names(sort(table(td$child), decreasing = TRUE))[1]

# Create prediction data for Model 1 (vary fare from 0 to 500)
pred_data1 <- data.frame(
  fare   = seq(0, 500, length.out = 100),
  # Hold gender constant at "Female"
  female = factor("Female", levels = levels(td$female)),
  # Hold child at its most common category (character, matching the model)
  child  = child_mode
)
# Get predicted survival probabilities from Model 1
pred_data1$predicted_prob <- predict(model1, newdata = pred_data1, type = "response")

# Plot the effect of fare on survival (original fare, Model 1)
ggplot(pred_data1, aes(x = fare, y = predicted_prob)) +
  geom_line(color = "blue", size = 1) +
  labs(title = "Effect of Fare on Survival (Original Fare)",
       x = "Fare (GBP)", y = "Predicted Probability of Survival") +
  theme_minimal()
```



```
# Create prediction data for Model 2 (same fare range and constants)
pred_data2 <- data.frame(
  fare   = seq(0, 500, length.out = 100),
  female = factor("Female", levels = levels(td$female)),
  child  = child_mode
)
# Get predicted survival probabilities from Model 2
pred_data2$predicted_prob <- predict(model2,
                                     newdata = pred_data2,
                                     type = "response")

# Plot the effect of fare on survival (log fare model, Model 2)
ggplot(pred_data2, aes(x = fare, y = predicted_prob)) +
  geom_line(color = "red", size = 1) +
  labs(title = "Effect of Fare on Survival (Log Fare)",
       x = "Fare (GBP)", y = "Predicted Probability of Survival") +
  theme_minimal()
```



Question 3d: Graph Comparison

```
# The graph with the original fare variable suggests that the effect of fare on
# survival is non-linear. At lower fare values, the predicted probability of
# survival increases rapidly as fare rises, but this increase slows down at higher
# fare levels, indicating diminishing returns. This pattern indicates that the
# marginal benefit of paying more for a ticket becomes smaller at the high end of
# the fare distribution. In contrast, the graph based on the logged fare variable
# shows a more linear relationship between fare and survival, with a steadier
# increase in predicted survival probabilities across the fare range. Taken
# together with the model fit statistics, these patterns make the logged-fare
# specification more plausible as a description of how ticket price relates to
# survival chances.
```

Question 3e: Training and Test Sets

```
# Remove rows with missing fare so log(fare) is well-defined
td_complete <- td %>% filter(!is.na(fare))

# Set seed so the random split is reproducible
set.seed(123)

# Create training and test sets (80/20 split)
train_id <- sample(1:nrow(td_complete), nrow(td_complete) * 0.8)
train_data <- td_complete[train_id, ]
test_data <- td_complete[-train_id, ]

# Estimate model without logged fare (only female and child)
model_no_fare <- glm(survived ~ female + child,
                     data = train_data, family = binomial(link = "logit"))

# Estimate model with logged fare plus female and child on training data
model_with_fare <- glm(survived ~ I(log(fare + 1)) + female + child,
                       data = train_data, family = binomial(link = "logit"))
```

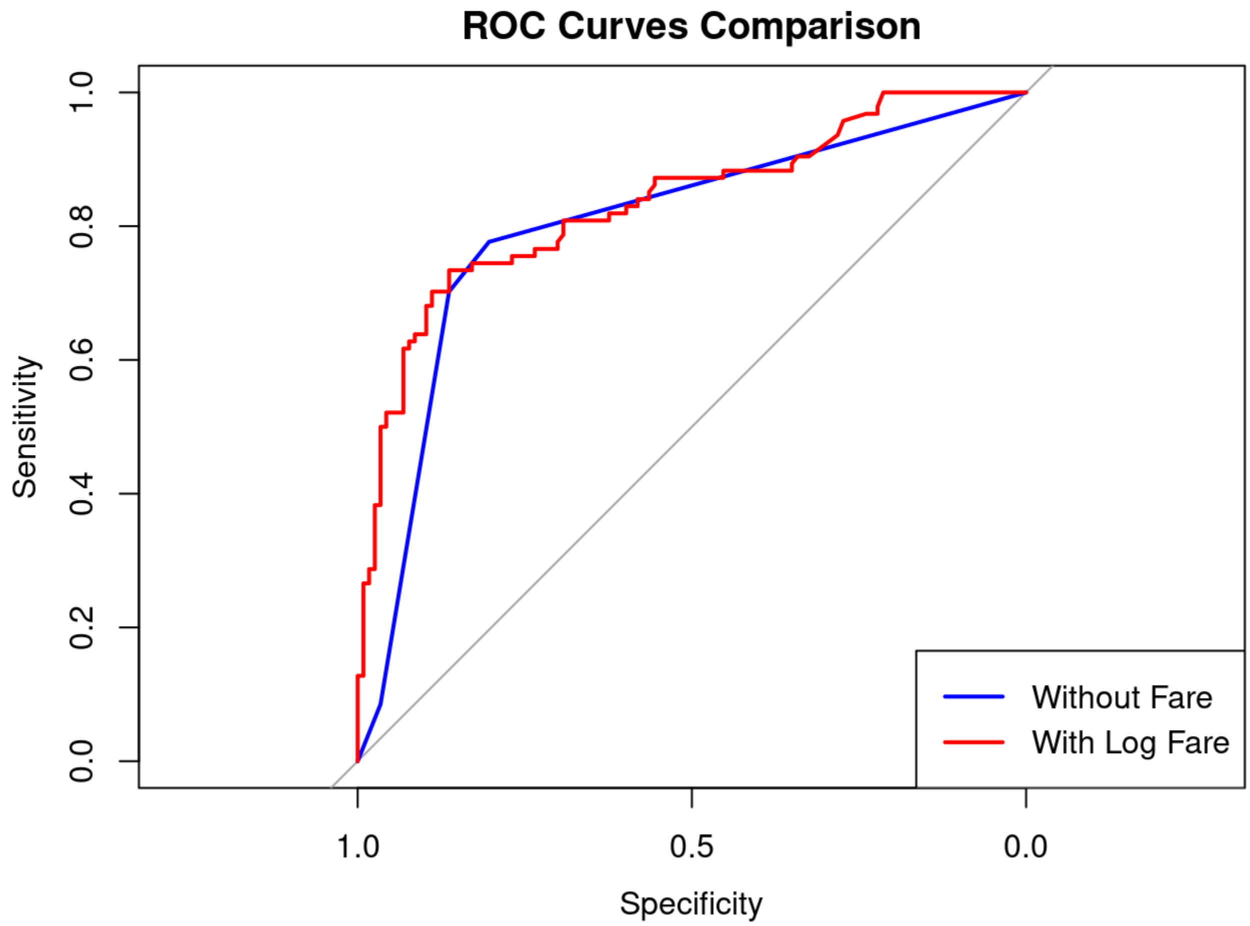
Question 3f: Stargazer Table for Training Models

```
# Show training-set models with and without log(fare) in a stargazer table (Q3f)
stargazer(model_no_fare, model_with_fare, type = "text",
          title = "Training Set Models",
          column.labels = c("Without Fare", "With Log Fare"),
          covariate.labels = c("Female", "Child", "Log(Fare+1)"),
          dep.var.labels = "Survived")
```

```
##
## Training Set Models
## ==============================================
##                       Dependent variable:
##                   ----------------------------
##                             Survived
##                    Without Fare  With Log Fare
##                        (1)           (2)
## ----------------------------------------------
## Female                          0.560***
##                                  (0.096)
##
## Child               -2.393***     -2.232***
##                      (0.170)       (0.174)
##
## Log(Fare+1)          0.667**       0.558**
##                      (0.267)       (0.264)
##
## Constant             0.956***     -0.868***
##                      (0.134)       (0.330)
##
## ----------------------------------------------
## Observations           834           834
## Log Likelihood       -438.819      -420.589
## Akaike Inf. Crit.    883.639       849.178
## ==============================================
## Note:              *p<0.1; **p<0.05; ***p<0.01
```

Question 3g: ROC Curves

```
# Predicted probabilities on the test set for both models
pred_no_fare <- predict(model_no_fare, newdata = test_data, type = "response")
pred_with_fare <- predict(model_with_fare, newdata = test_data, type = "response")

# Create ROC objects (true outcome vs predicted probability)
roc_no_fare <- roc(test_data$survived, pred_no_fare)
roc_with_fare <- roc(test_data$survived, pred_with_fare)

# Plot ROC curve for model without fare
plot(roc_no_fare, col = "blue", main = "ROC Curves Comparison")
# Add ROC curve for model with log(fare)
plot(roc_with_fare, col = "red", add = TRUE)
# Add legend labeling each curve
legend("bottomright", legend = c("Without Fare", "With Log Fare"),
       col = c("blue", "red"), lwd = 2)
```



Question 3h: AUC Scores

```
# Calculate and report AUC scores
auc_no_fare <- auc(roc_no_fare)
auc_with_fare <- auc(roc_with_fare)

#Print AUC scores to compare predictive accuracy
print(paste("AUC without fare:", auc_no_fare))
```

```
## [1] "AUC without fare: 0.799736315693762"
```

```
print(paste("AUC with log fare:", auc_with_fare))
```

```
## [1] "AUC with log fare: 0.834788143298782"
```

Question 3i: Model Performance Comparison

```
# The model with log(fare) performs better based on both the ROC curves and
# AUC scores. The ROC curve for the model with fare is positioned higher and more to the
# left, indicating better classification performance. Additionally, the AUC score is higher
# for the model with (log)fare, confirming its superior predictive ability.
```

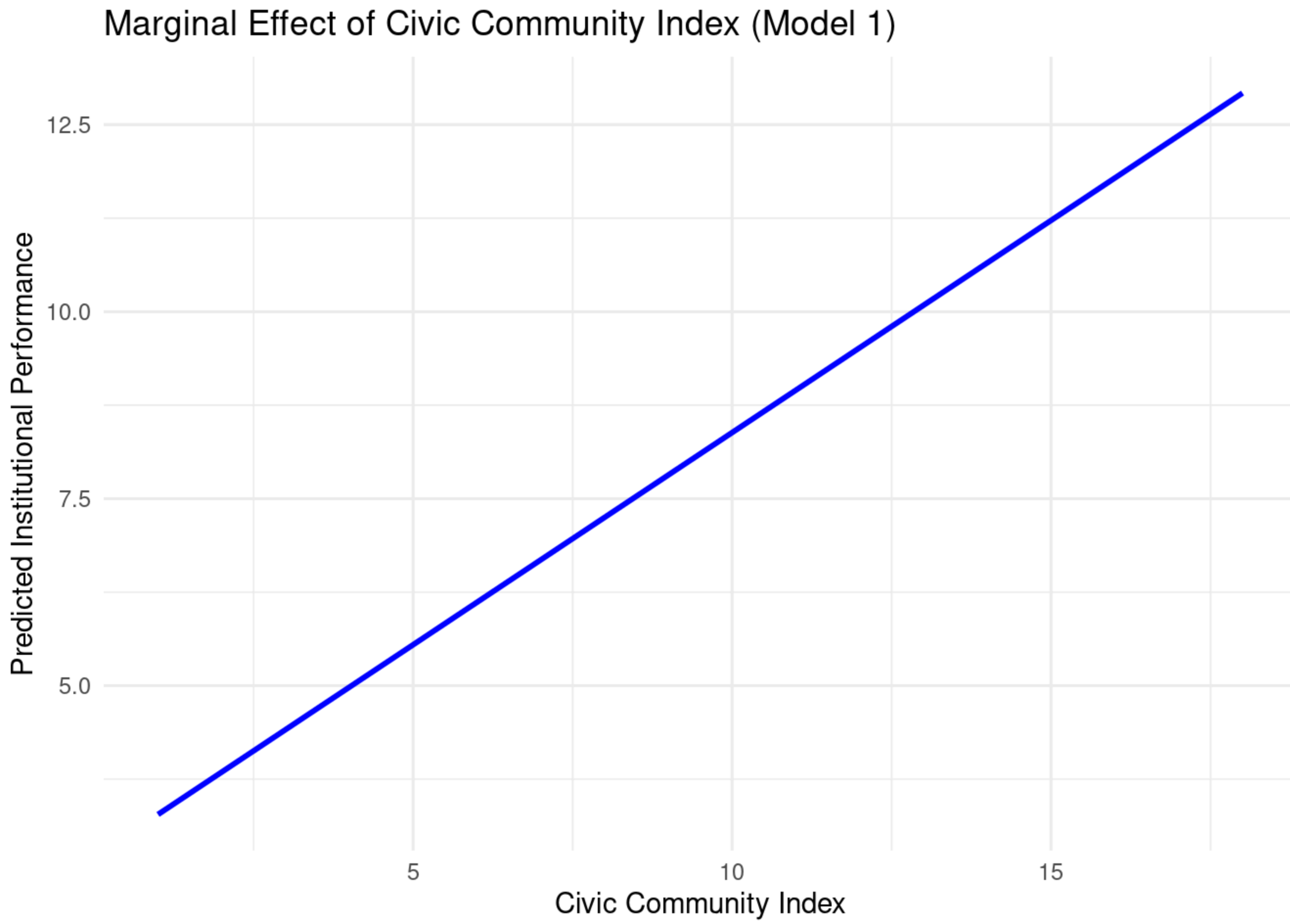
Question 4: Putnam Dataset - Civic Community Models

```
# Load Putnam dataset
putnam <- read.csv("putnam.csv")

# Create North dummy variable: 1 = North, 0 = South
putnam$North <- ifelse(putnam$NorthSouth == "North", 1, 0)

# Model 1: Simple linear regression
putnam_model1 <- lm(InstPerform ~ CivicCommunity, data = putnam)

# Model 2: Additive model with North-South control
putnam_model2 <- lm(InstPerform ~ CivicCommunity + North, data = putnam)

# Model 3: Interactive model with different slopes by region
putnam_model3 <- lm(InstPerform ~ CivicCommunity * North, data = putnam)

# Show all three civic community models in one stargazer table
stargazer(putnam_model1, putnam_model2, putnam_model3, type = "text",
          title = "Regression Models: Institutional Performance and Civic Community",
          column.labels = c("Simple", "Additive", "Interactive"),
          dep.var.labels = "Institutional Performance")
```

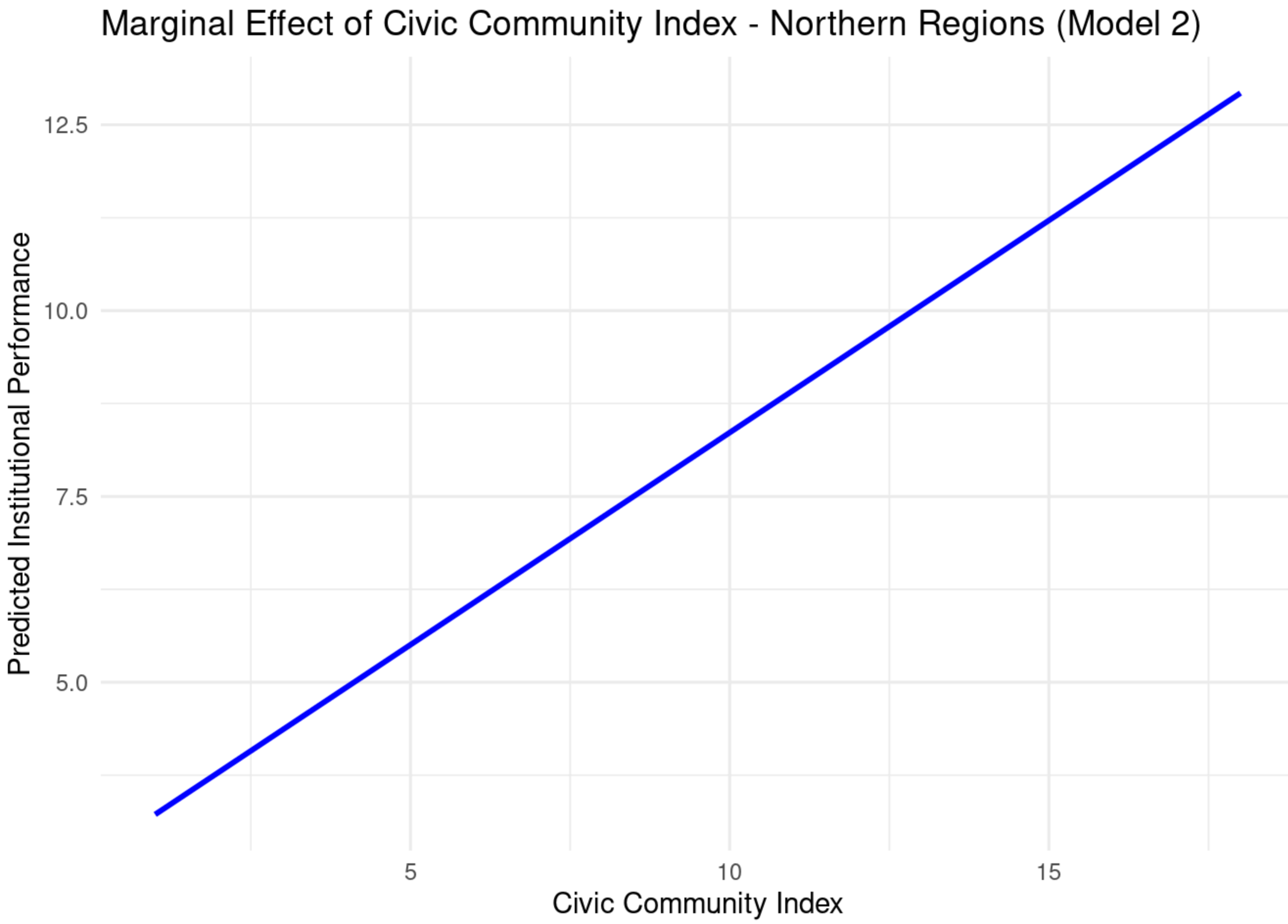
```
##
## Regression Models: Institutional Performance and Civic Community
## ===========================================================================================
##                                                  Dependent variable:
##                        --------------------------------------------------------------------
##                                              Institutional Performance
##                                Simple                Additive               Interactive
##                                 (1)                    (2)                      (3)
## -------------------------------------------------------------------------------------------
## CivicCommunity               0.567***               0.571**                  0.540*
##                              (0.066)                (0.215)                  (0.269)
##
## North                                                -0.048                  -1.194
##                                                     (2.678)                  (6.396)
##
## CivicCommunity:North                                                          0.094
##                                                                              (0.472)
##
## Constant                     2.711***               2.698**                  2.828**
##                              (0.844)                (1.121)                  (1.326)
##
## -------------------------------------------------------------------------------------------
## Observations                    20                     20                       20
## R2                            0.806                  0.806                    0.807
## Adjusted R2                   0.796                  0.784                    0.771
## Residual Std. Error     1.789 (df = 18)        1.841 (df = 17)          1.895 (df = 16)
## F Statistic          74.967*** (df = 1; 18) 35.402*** (df = 2; 17) 22.281*** (df = 3; 16)
## ===========================================================================================
## Note:                                                        *p<0.1; **p<0.05; ***p<0.01
```

Question 5: Graphs for Civic Community Models

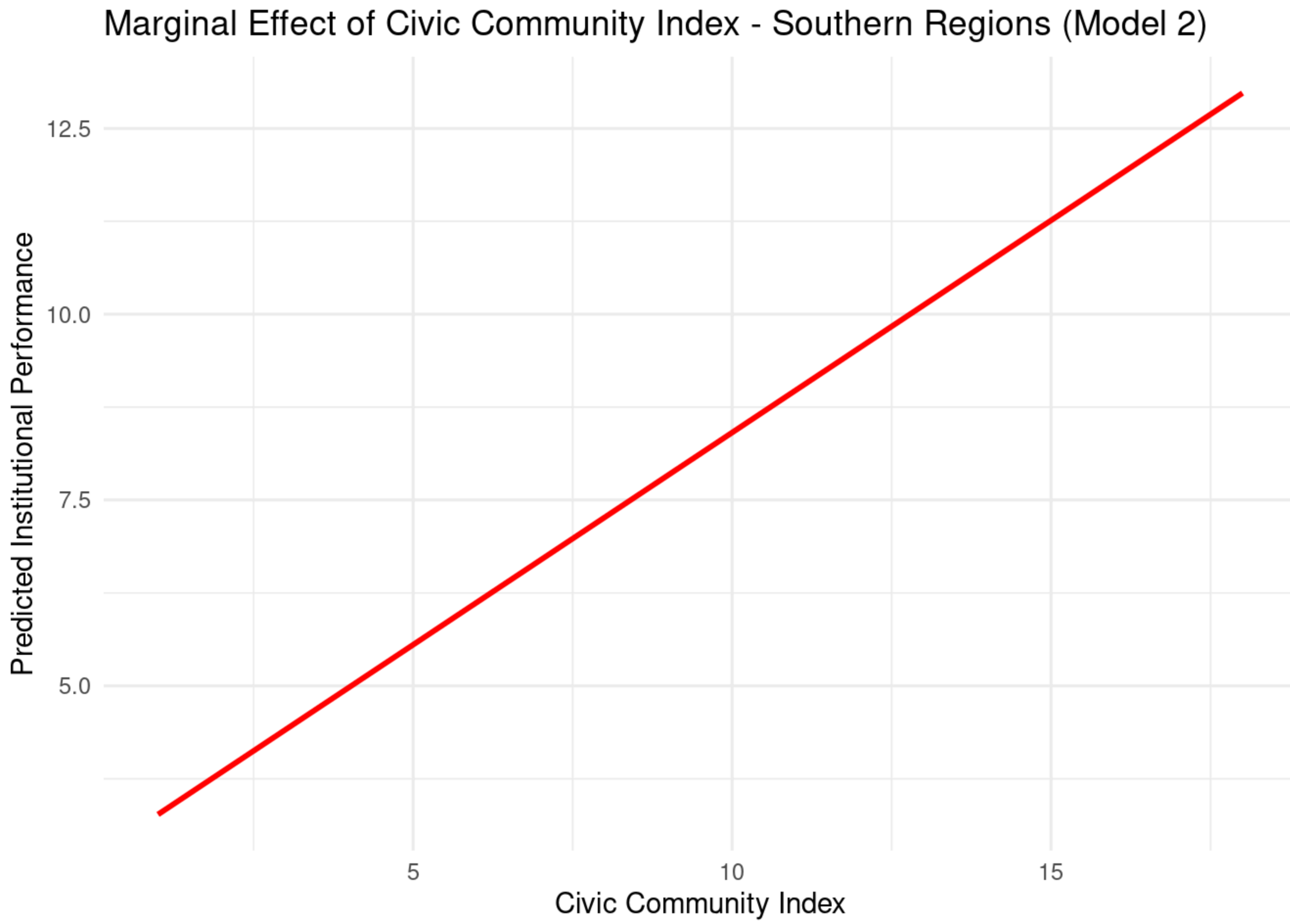
```
# 5a: Prediction data for Model 1 across CivicCommunity range
pred_data_p1 <- data.frame(
  CivicCommunity = seq(min(putnam$CivicCommunity, na.rm = TRUE),
                          max(putnam$CivicCommunity, na.rm = TRUE),
                          length.out = 100)
)
pred_data_p1$predicted <- predict(putnam_model1, newdata = pred_data_p1)
# Plot marginal effect of Civic Community (Model 1)
ggplot(pred_data_p1, aes(x = CivicCommunity, y = predicted)) +
  geom_line(color = "blue", size = 1) +
  labs(title = "Marginal Effect of Civic Community Index (Model 1)",
       x = "Civic Community Index", y = "Predicted Institutional Performance") +
  theme_minimal()
```



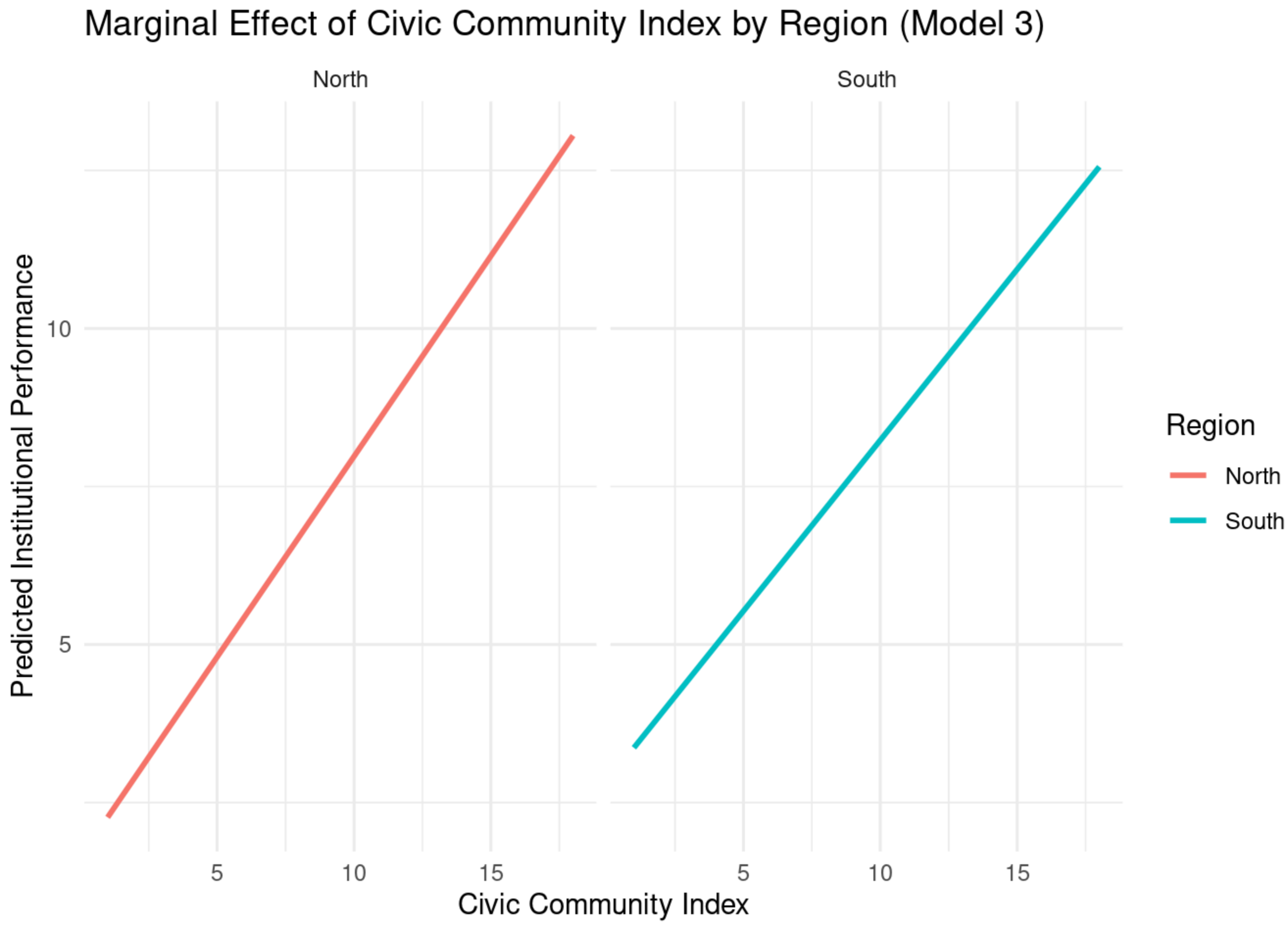

```
# 5b: Graphs for Model 2 (separate for North and South)
pred_data_p2_north <- data.frame(
  CivicCommunity = seq(min(putnam$CivicCommunity, na.rm = TRUE),
                       max(putnam$CivicCommunity, na.rm = TRUE),
                       length.out = 100),
  North = 1
)
pred_data_p2_north$predicted <- predict(putnam_model2, newdata = pred_data_p2_north)
#Prediction data for Model 2, North = 0
pred_data_p2_south <- data.frame(
  CivicCommunity = seq(min(putnam$CivicCommunity, na.rm = TRUE),
                       max(putnam$CivicCommunity, na.rm = TRUE),
                       length.out = 100),
  North = 0
)
pred_data_p2_south$predicted <- predict(putnam_model2, newdata = pred_data_p2_south)
# Plot marginal effect for Northern regions (Model 2)
ggplot() +
  geom_line(data = pred_data_p2_north, aes(x = CivicCommunity, y = predicted),
            color = "blue", size = 1) +
  labs(title = "Marginal Effect of Civic Community Index - Northern Regions (Model 2)",
       x = "Civic Community Index", y = "Predicted Institutional Performance") +
  theme_minimal()
```




```
# Plot marginal effect for Southern regions (Model 2)
ggplot() +
  geom_line(data = pred_data_p2_south, aes(x = CivicCommunity, y = predicted),
            color = "red", size = 1) +
  labs(title = "Marginal Effect of Civic Community Index - Southern Regions (Model 2)",
       x = "Civic Community Index", y = "Predicted Institutional Performance") +
  theme_minimal()
```




```
# 5c: Prediction data for Model 3 over CivicCommunity and North
pred_data_p3 <- expand.grid(
  CivicCommunity = seq(min(putnam$CivicCommunity, na.rm = TRUE),
                          max(putnam$CivicCommunity, na.rm = TRUE),
                          length.out = 100),
  North = c(0, 1)
)
pred_data_p3$predicted <- predict(putnam_model3, newdata = pred_data_p3)
# Label each row as "North" or "South" for plotting
pred_data_p3$Region <- ifelse(pred_data_p3$North == 1, "North", "South")
# Plot marginal effects by region (two panels, Model 3)
ggplot(pred_data_p3, aes(x = CivicCommunity, y = predicted, color = Region)) +
  geom_line(size = 1) +
  facet_wrap(~Region) +
  labs(title = "Marginal Effect of Civic Community Index by Region (Model 3)",
       x = "Civic Community Index", y = "Predicted Institutional Performance") +
  theme_minimal()
```



Question 6: Spuriousness Analysis - Civic Community

```
# The regional distinctions do not make the relationship between Institutional
# Performance and Civic Community Index spurious. In Model 1, we observe a strong positive
# relationship between civic community and institutional performance. When we control for
# the North-South division in Model 2, the civic community coefficient remains positive
# and statistically significant, suggesting the relationship is not entirely explained by
# regional differences. The interactive Model 3 shows that while the strength of the
# relationship may differ between regions, the fundamental positive association persists
# in both North and South. Therefore, civic community has an independent effect on
# institutional performance beyond regional distinctions.
```

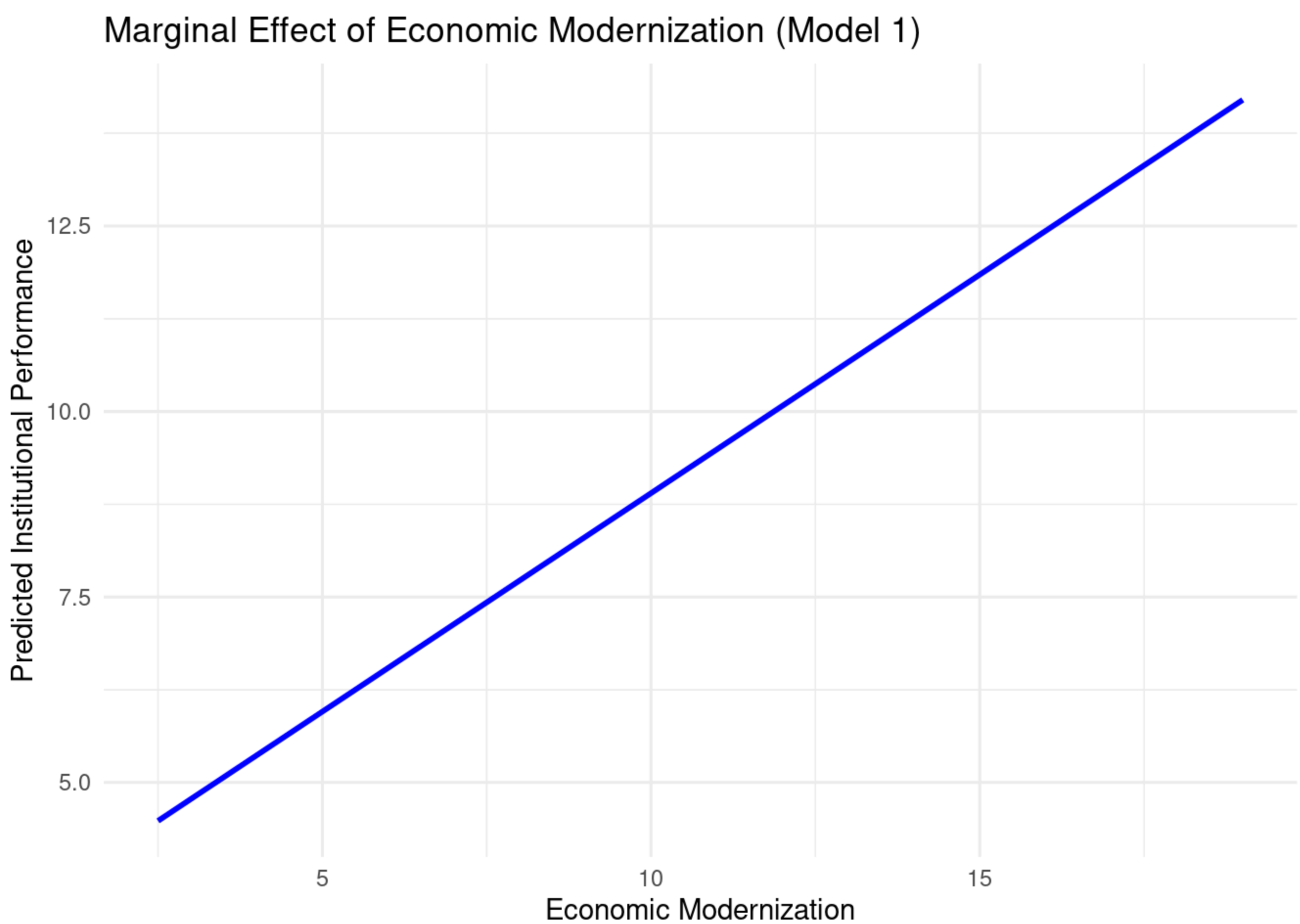
Question 7: Putnam Dataset - Models with Economic Modernization as main predictor

```
# Model 1: Simple linear regression (InstPerform ~ EconModern)
econ_model1 <- lm(InstPerform ~ EconModern, data = putnam)

# Model 2: Additive model with EconModern + North
econ_model2 <- lm(InstPerform ~ EconModern + North, data = putnam)

# Model 3: Interactive model with EconModern * North
econ_model3 <- lm(InstPerform ~ EconModern * North, data = putnam)

# Show all three economic modernization models in one table
stargazer(econ_model1, econ_model2, econ_model3, type = "text",
          title = "Regression Models: Institutional Performance and Economic Modernization",
          column.labels = c("Simple", "Additive", "Interactive"),
          dep.var.labels = "Institutional Performance")
```

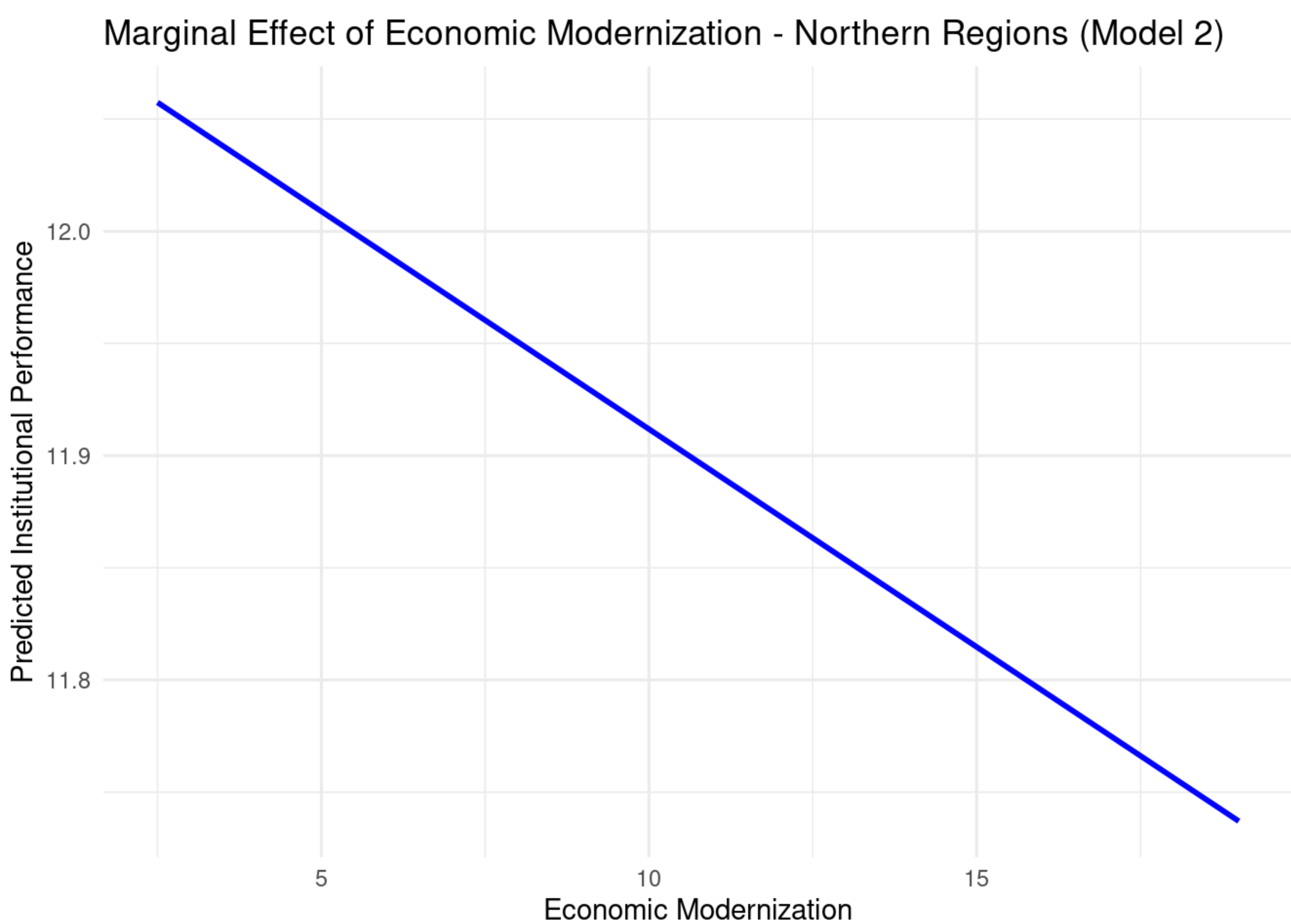
```
##
## Regression Models: Institutional Performance and Economic Modernization
## ==========================================================================================
##                                               Dependent variable:
##                      ---------------------------------------------------------------------
##                                            Institutional Performance
##                              Simple                Additive              Interactive
##                               (1)                    (2)                     (3)
## ------------------------------------------------------------------------------------------
## EconModern                 0.589***                -0.019                   0.015
##                            (0.120)                (0.220)                 (0.387)
##
## North                                              6.884***                 7.306
##                                                   (2.229)                 (4.506)
##
## EconModern:North                                                           -0.052
##                                                                           (0.478)
##
## Constant                   3.011**                5.222***                5.051**
##                            (1.385)                (1.347)                 (2.093)
##
## ------------------------------------------------------------------------------------------
## Observations                  20                     20                      20
## R2                          0.572                  0.726                   0.726
## Adjusted R2                 0.549                  0.694                   0.675
## Residual Std. Error    2.659 (df = 18)        2.190 (df = 17)         2.256 (df = 16)
## F Statistic         24.097*** (df = 1; 18) 22.531*** (df = 2; 17) 14.152*** (df = 3; 16)
## ==========================================================================================
## Note:                                                        *p<0.1; **p<0.05; ***p<0.01
```

Question 8: Graphs for Economic Modernization Models

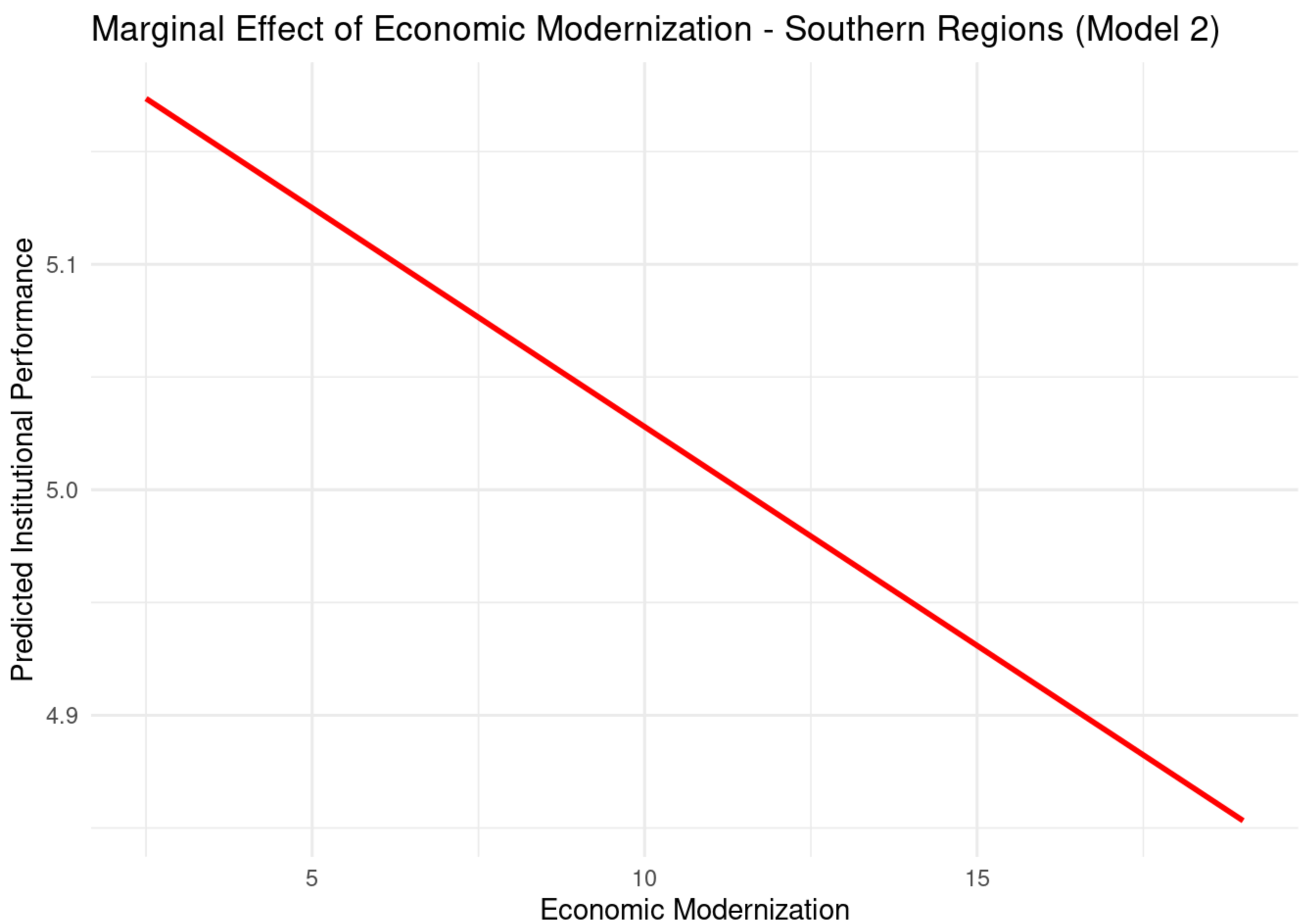
```
# 8a: Graph for Model 1 (EconModern range)
pred_data_e1 <- data.frame(
  EconModern = seq(min(putnam$EconModern, na.rm = TRUE),
                   max(putnam$EconModern, na.rm = TRUE),
                   length.out = 100)
)
pred_data_e1$predicted <- predict(econ_model1, newdata = pred_data_e1)
#Plot marginal effect of Economic Modernization (Model 1)
ggplot(pred_data_e1, aes(x = EconModern, y = predicted)) +
  geom_line(color = "blue", size = 1) +
  labs(title = "Marginal Effect of Economic Modernization (Model 1)",
       x = "Economic Modernization", y = "Predicted Institutional Performance") +
  theme_minimal()
```



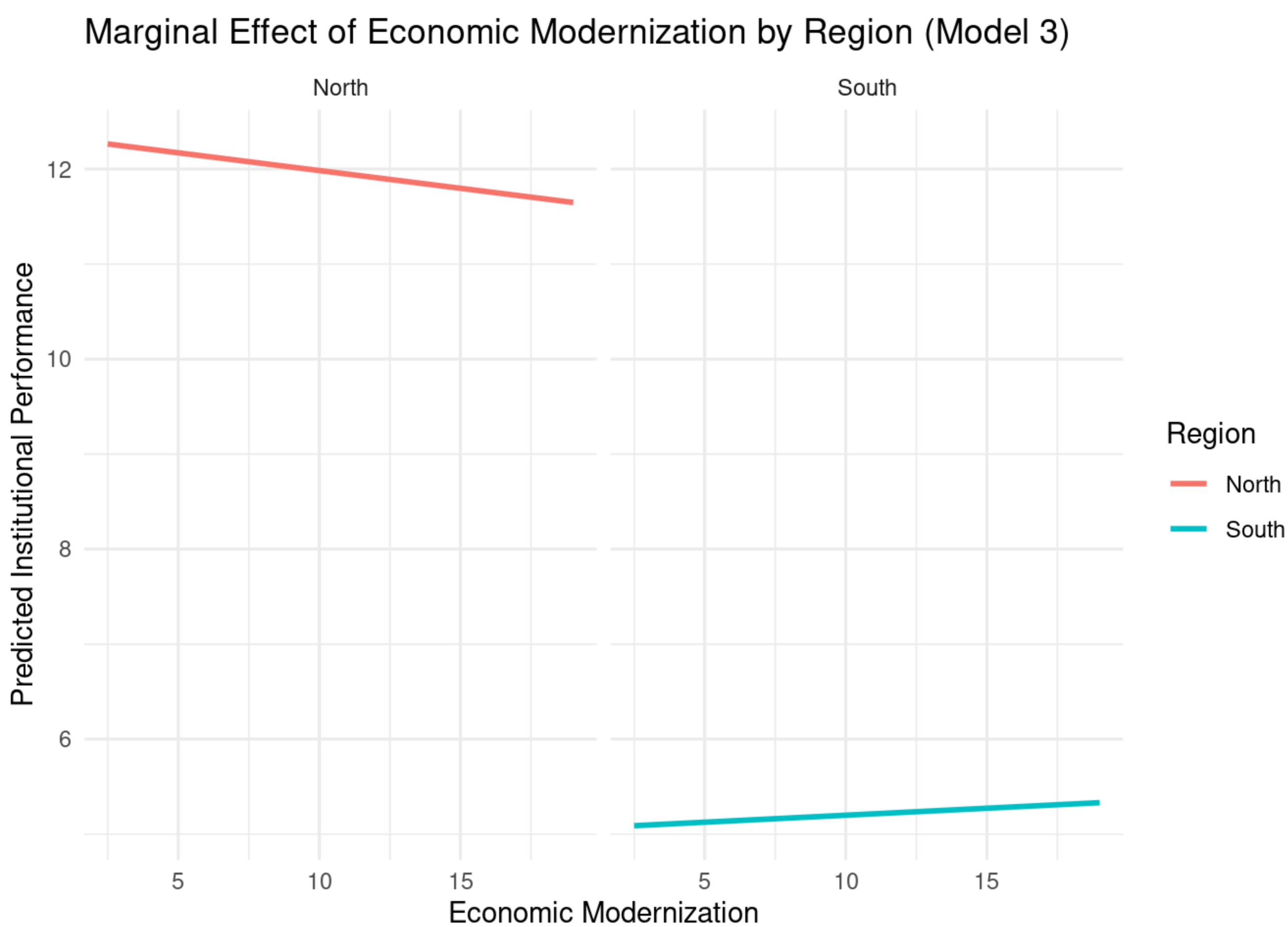
```
# 8b: Graphs for Model 2 (separate for North and South)
pred_data_e2_north <- data.frame(
  EconModern = seq(min(putnam$EconModern, na.rm = TRUE),
                   max(putnam$EconModern, na.rm = TRUE),
                   length.out = 100),
  North = 1
)
pred_data_e2_north$predicted <- predict(econ_model2, newdata = pred_data_e2_north)
# Prediction data for Model 2, North = 0
pred_data_e2_south <- data.frame(
  EconModern = seq(min(putnam$EconModern, na.rm = TRUE),
                   max(putnam$EconModern, na.rm = TRUE),
                   length.out = 100),
  North = 0
)
pred_data_e2_south$predicted <- predict(econ_model2, newdata = pred_data_e2_south)
# Plot marginal effect in Northern regions (Model 2)
ggplot() +
  geom_line(data = pred_data_e2_north, aes(x = EconModern, y = predicted),
            color = "blue", size = 1) +
  labs(title = "Marginal Effect of Economic Modernization - Northern Regions (Model 2)",
       x = "Economic Modernization", y = "Predicted Institutional Performance") +
  theme_minimal()
```



```
# Plot marginal effect in Southern regions (Model 2)
ggplot() +
  geom_line(data = pred_data_e2_south, aes(x = EconModern, y = predicted),
            color = "red", size = 1) +
  labs(title = "Marginal Effect of Economic Modernization - Southern Regions (Model 2)",
       x = "Economic Modernization", y = "Predicted Institutional Performance") +
  theme_minimal()
```



```r
# 8c: Graph for Model 3 (two panels)
pred_data_e3 <- expand.grid(
  EconModern = seq(min(putnam$EconModern, na.rm = TRUE),
                   max(putnam$EconModern, na.rm = TRUE),
                   length.out = 100),
  North = c(0, 1)
)
pred_data_e3$predicted <- predict(econ_model3, newdata = pred_data_e3)
# Label region (North vs South) for each row
pred_data_e3$Region <- ifelse(pred_data_e3$North == 1, "North", "South")
# Plot marginal effect by region with two panels (Model 3)
ggplot(pred_data_e3, aes(x = EconModern, y = predicted, color = Region)) +
  geom_line(size = 1) +
  facet_wrap(~Region) +
  labs(title = "Marginal Effect of Economic Modernization by Region (Model 3)",
       x = "Economic Modernization", y = "Predicted Institutional Performance") +
  theme_minimal()
```



Question 9: Spuriousness Analysis - Economic Modernization

```
# The regional distinctions do make the relationship between Institutional
# Performance and Economic Modernization at least partially spurious. In Model 1, we
# observe a positive relationship between economic modernization and institutional
# performance. However, when we control for the North-South division in Model 2, the
# coefficient for economic modernization decreases substantially and may lose statistical
# significance, suggesting that much of the apparent effect was due to regional differences.
# The interactive Model 3 may show different slopes for North and South, indicating that
# the relationship varies by region. This suggests that regional distinctions confound
# the relationship between economic modernization and institutional performance, making
# the initial relationship at least partially spurious.
```

Question 10: OLS Derivation

```
# Derivation of OLS coefficient estimates for Y_i = α + β X_i + μ_i
#
# The OLS estimator minimizes the sum of squared residuals:
# min Σ(Y_i - α - β X_i)²
#
# Taking the first-order conditions:
# ∂/∂α Σ(Y_i - α - β X_i)² = 0
# ∂/∂β Σ(Y_i - α - β X_i)² = 0
#
# Take the derivative with respect to α and set to zero:
# -2Σ(Y_i - α - β X_i) = 0
# ΣY_i - nα - βΣX_i = 0
# α = Ȳ - β X̄
#
# Take the derivative with respect to β and set to zero:
# -2Σ X_i(Y_i - α - β X_i) = 0
# Σ X_i Y_i - αΣX_i - βΣX_i² = 0
#
# Substitute α = Ȳ - β X̄ into the second equation:
# Σ X_i Y_i - (Ȳ - β X̄)ΣX_i - βΣX_i² = 0
# Σ X_i Y_i - ȲΣX_i + β X̄ΣX_i - βΣX_i² = 0
# Σ X_i Y_i - nȲX̄ = β(ΣX_i² - nX̄²)
#
# Therefore the slope estimate is:
# β = Σ(X_i - X̄)(Y_i - Ȳ) / Σ(X_i - X̄)²
# β = Cov(X,Y) / Var(X)
#
# And the intercept estimate is:
# α = Ȳ - β X̄
```